물리화학2 (2011년 2학기) 3차 과제 해답 (**총 75점**)

2011년 11월 11일 (금요일)

1. **(총 15점, 각 5점)**

(a) 문제로부터 X 와 x 가 다음과 같이 정의되므로:

$$X = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}, \tag{1}$$

$$x = x_2 - x_1 - x_{eq}, \tag{2}$$

일차 편도함수는 다음과 같이 얻어진다.

$$\left(\frac{\partial X}{\partial x_1}\right)_{x_2} = \frac{m_1}{m_1 + m_2}, \tag{3}$$

$$\left(\frac{\partial x}{\partial x_1}\right)_{x_2} = -1, \tag{4}$$

$$\left(\frac{\partial X}{\partial x_2}\right)_{x_1} = \frac{m_2}{m_2 + m_2}, \tag{5}$$

$$\left(\frac{\partial x}{\partial x_2}\right)_{x_1} = 1. \tag{6}$$

연쇄법칙을 이용하면 다음과 같은 결과를 얻는다.

$$\boxed{\begin{aligned} \frac{\partial}{\partial x_1} &= \frac{\partial X}{\partial x_1}\frac{\partial}{\partial X} + \frac{\partial x}{\partial x_1}\frac{\partial}{\partial x}, & (7)\\ &= \frac{m_1}{m_1 + m_2}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}, & (8) \end{aligned}}$$

그리고,

$$\boxed{\begin{aligned} \frac{\partial}{\partial x_2} &= \frac{\partial X}{\partial x_2}\frac{\partial}{\partial X} + \frac{\partial x}{\partial x_2}\frac{\partial}{\partial x}, & (9)\\ &= \frac{m_2}{m_2 + m_2}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}. & (10) \end{aligned}}$$

(b) (8) 식으로부터,

$$\begin{aligned} \frac{\partial^2}{\partial x_1^2} &= \frac{\partial}{\partial x_1}\cdot\frac{\partial}{\partial x_1}, \\ &= \frac{\partial}{\partial x_1}\left(\frac{m_1}{m_1 + m_2}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}\right). \end{aligned} \tag{11}$$

g 를 다음과 같이 정의한다면,

$$g \equiv \frac{m_1}{m_1 + m_2}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}, \tag{12}$$

(11) 식은 다음과 같이 쓸 수 있다.

$$\begin{aligned}
\frac{\partial^2}{\partial x_1^2} &= \frac{\partial}{\partial x_1}\cdot\frac{\partial}{\partial x_1}, \\
&= \frac{\partial}{\partial x_1}\left(\frac{m_1}{m_1+m_2}\frac{\partial}{\partial X}-\frac{\partial}{\partial x}\right), \\
&= \frac{\partial g}{\partial x_1}, \\
&= \frac{\partial X}{\partial x_1}\frac{\partial g}{\partial X}+\frac{\partial x}{\partial x_1}\frac{\partial g}{\partial x}, \qquad (\text{ (7) 식으로부터}) \\
&= \frac{m_1}{m_1+m_2}\frac{\partial g}{\partial X}-\frac{\partial g}{\partial x}, \qquad (\text{(3) 식과 (4) 식으로부터}) \\
&= \frac{m_1}{m_1+m_2}\frac{\partial}{\partial X}\left(\frac{m_1}{m_1+m_2}\frac{\partial}{\partial X}-\frac{\partial}{\partial x}\right)-\frac{\partial}{\partial x}\left(\frac{m_1}{m_1+m_2}\frac{\partial}{\partial X}-\frac{\partial}{\partial x}\right), \\
&= \boxed{\frac{m_1^2}{(m_1+m_2)^2}\frac{\partial^2}{\partial X^2}-\frac{2m_1}{m_1+m_2}\frac{\partial^2}{\partial X\partial x}+\frac{\partial^2}{\partial x^2}.} \qquad (13)
\end{aligned}$$

같은 방법으로 (10) 식으로부터,

$$\begin{aligned}
\frac{\partial^2}{\partial x_2^2} &= \frac{\partial}{\partial x_2}\cdot\frac{\partial}{\partial x_2}, \\
&= \frac{\partial}{\partial x_2}\left(\frac{m_2}{m_1+m_2}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}\right). \qquad (14)
\end{aligned}$$

h를 다음과 같이 정의한다면,

$$h \equiv \frac{m_2}{m_1+m_2}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}, \qquad (15)$$

(14) 식은 다음과 같이 쓸 수 있다.

$$\begin{aligned}
\frac{\partial^2}{\partial x_2^2} &= \frac{\partial}{\partial x_2}\cdot\frac{\partial}{\partial x_2}, \\
&= \frac{\partial}{\partial x_2}\left(\frac{m_2}{m_1+m_2}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}\right), \\
&= \frac{\partial h}{\partial x_2}, \\
&= \frac{\partial X}{\partial x_2}\frac{\partial h}{\partial X}+\frac{\partial x}{\partial x_2}\frac{\partial h}{\partial x}, \qquad (\text{ (9) 식으로부터 }) \\
&= \frac{m_2}{m_1+m_2}\frac{\partial h}{\partial X}+\frac{\partial h}{\partial x}, \qquad (\text{(5) 식과 (6) 식으로부터}) \\
&= \frac{m_2}{m_1+m_2}\frac{\partial}{\partial X}\left(\frac{m_2}{m_1+m_2}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}\right)+\frac{\partial}{\partial x}\left(\frac{m_2}{m_1+m_2}\frac{\partial}{\partial X}+\frac{\partial}{\partial x}\right), \\
&= \boxed{\frac{m_2^2}{(m_1+m_2)^2}\frac{\partial^2}{\partial X^2}+\frac{2m_2}{m_1+m_2}\frac{\partial^2}{\partial X\partial x}+\frac{\partial^2}{\partial x^2}.} \qquad (16)
\end{aligned}$$

(c) (13) 식과 (16) 식으로부터,

$$\begin{aligned}
\frac{1}{m_1}\frac{\partial^2}{\partial x_1^2}+\frac{1}{m_2}\frac{\partial^2}{\partial x_2^2} &= \frac{1}{m_1}\left[\frac{m_1^2}{(m_1+m_2)^2}\frac{\partial^2}{\partial X^2}-\frac{2m_1}{m_1+m_2}\frac{\partial^2}{\partial X\partial x}+\frac{\partial^2}{\partial x^2}\right] \\
&\quad +\frac{1}{m_2}\left[\frac{m_2^2}{(m_1+m_2)^2}\frac{\partial^2}{\partial X^2}+\frac{2m_2}{m_1+m_2}\frac{\partial^2}{\partial X\partial x}+\frac{\partial^2}{\partial x^2}\right] \\
&= \frac{m_1}{(m_1+m_2)^2}\frac{\partial^2}{\partial X^2}+\frac{2}{m_1+m_2}\frac{\partial^2}{\partial X\partial x}+\frac{1}{m_1}\frac{\partial^2}{\partial x^2} \\
&\quad +\frac{m_2}{(m_1+m_2)^2}\frac{\partial^2}{\partial X^2}-\frac{2}{m_1+m_2}\frac{\partial^2}{\partial X\partial x}+\frac{1}{m_2}\frac{\partial^2}{\partial x^2} \\
&= \boxed{\frac{1}{m_1+m_2}\frac{\partial^2}{\partial X^2}+\left(\frac{1}{m_1}+\frac{1}{m_2}\right)\frac{\partial^2}{\partial x^2}} \qquad (17)
\end{aligned}$$

2. **(총 15 점, 각 5 점)**

$$x=r\cos\phi,\quad y=r\sin\phi\Rightarrow r^2=x^2+y^2,\quad \tan\phi=\frac{y}{x}. \qquad (18)$$

(a) 연쇄법칙을 이용하면,

$$\left(\frac{\partial}{\partial x}\right)_y = \left(\frac{\partial r}{\partial x}\right)_y\left(\frac{\partial}{\partial r}\right)_\phi+\left(\frac{\partial \phi}{\partial x}\right)_y\left(\frac{\partial}{\partial \phi}\right)_r, \qquad (19a)$$

$$\left(\frac{\partial}{\partial y}\right)_x = \left(\frac{\partial r}{\partial y}\right)_x\left(\frac{\partial}{\partial r}\right)_\phi+\left(\frac{\partial \phi}{\partial y}\right)_x\left(\frac{\partial}{\partial \phi}\right)_r, \qquad (19b)$$

이고,

$$\frac{\partial r}{\partial x}=\frac{x}{r}=\cos\phi,\quad \frac{\partial\phi}{\partial x}=-\frac{\sin\phi}{r}, \qquad (20a)$$

$$\frac{\partial r}{\partial y}=\frac{y}{r}=\sin\phi,\quad \frac{\partial\phi}{\partial y}=\frac{\cos\phi}{r}, \qquad (20b)$$

이므로,

$$\boxed{\begin{aligned}
\frac{\partial}{\partial x} &= \cos\phi\frac{\partial}{\partial r}-\frac{\sin\phi}{r}\frac{\partial}{\partial\phi}, \qquad (21a)\\
\frac{\partial}{\partial y} &= \sin\phi\frac{\partial}{\partial r}+\frac{\cos\phi}{r}\frac{\partial}{\partial\phi}. \qquad (21b)
\end{aligned}}$$

(b) g를 다음과 같이 정의하면,

$$g\equiv\frac{\partial}{\partial x}=\cos\phi\frac{\partial}{\partial r}-\frac{\sin\phi}{r}\frac{\partial}{\partial\phi}, \qquad (22)$$

$\partial^2/\partial x^2$ 는 다음과 같이 쓸 수 있다.

$$\begin{aligned}
\frac{\partial^2}{\partial x^2} &= \frac{\partial}{\partial x}\cdot\frac{\partial}{\partial x}, \\
&= \frac{\partial g}{\partial x}, \\
&= \frac{\partial r}{\partial x}\frac{\partial g}{\partial r}+\frac{\partial \phi}{\partial x}\frac{\partial g}{\partial \phi}, \qquad (\text{ (19a) 식으로부터}) \\
&= \cos\phi\frac{\partial g}{\partial r}-\frac{\sin\phi}{r}\frac{\partial g}{\partial \phi}, \qquad (\text{(20a) 식과 (20b) 식으로부터}) \\
&= \cos\phi\frac{\partial}{\partial r}\left(\cos\phi\frac{\partial}{\partial r}-\frac{\sin\phi}{r}\frac{\partial}{\partial \phi}\right)-\frac{\sin\phi}{r}\frac{\partial}{\partial \phi}\left(\cos\phi\frac{\partial}{\partial r}-\frac{\sin\phi}{r}\frac{\partial}{\partial \phi}\right), \\
&= \cos^2\phi\frac{\partial^2}{\partial r^2}+\frac{\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial \phi}-\frac{\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial r\partial \phi} \\
&\quad +\frac{\sin^2\phi}{r}\frac{\partial}{\partial r}-\frac{\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial \phi\partial r}+\frac{\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial \phi}+\frac{\sin^2\phi}{r^2}\frac{\partial^2}{\partial \phi^2}. \\
&= \boxed{\cos^2\phi\frac{\partial^2}{\partial r^2}+\frac{2\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial \phi}-\frac{2\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial r\partial \phi}+\frac{\sin^2\phi}{r}\frac{\partial}{\partial r}+\frac{\sin^2\phi}{r^2}\frac{\partial^2}{\partial \phi^2}.}
\end{aligned} \tag{23}$$

같은 방식으로 h 를 다음과 같이 정의하면

$$h \equiv \frac{\partial}{\partial y} = \sin\phi\frac{\partial}{\partial r}+\frac{\cos\phi}{r}\frac{\partial}{\partial \phi}, \tag{24}$$

$\partial^2/\partial y^2$ 는 다음과 같이 쓸 수 있다.

$$\begin{aligned}
\frac{\partial^2}{\partial y^2} &= \frac{\partial}{\partial y}\cdot\frac{\partial}{\partial y}, \\
&= \frac{\partial h}{\partial y}, \\
&= \frac{\partial r}{\partial y}\frac{\partial h}{\partial r}+\frac{\partial \phi}{\partial y}\frac{\partial h}{\partial \phi}, \qquad (\text{(19b) 식으로부터}) \\
&= \sin\phi\frac{\partial h}{\partial r}+\frac{\cos\phi}{r}\frac{\partial h}{\partial \phi}, \qquad (\text{(20a) 식과 (20b) 식으로부터}) \\
&= \sin\phi\frac{\partial}{\partial r}\left(\sin\phi\frac{\partial}{\partial r}+\frac{\cos\phi}{r}\frac{\partial}{\partial \phi}\right)+\frac{\cos\phi}{r}\frac{\partial}{\partial \phi}\left(\sin\phi\frac{\partial}{\partial r}+\frac{\cos\phi}{r}\frac{\partial}{\partial \phi}\right), \\
&= \sin^2\phi\frac{\partial^2}{\partial r^2}-\frac{\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial \phi}+\frac{\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial r\partial \phi} \\
&\quad +\frac{\cos^2\phi}{r}\frac{\partial}{\partial r}+\frac{\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial \phi\partial r}-\frac{\cos\phi\sin\phi}{r^2}\frac{\partial}{\partial \phi}+\frac{\cos^2\phi}{r^2}\frac{\partial^2}{\partial \phi^2}. \\
&= \boxed{\sin^2\phi\frac{\partial^2}{\partial r^2}-\frac{2\cos\phi\sin\phi}{r^2}\frac{\partial}{\partial \phi}+\frac{2\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial r\partial \phi}+\frac{\cos^2\phi}{r}\frac{\partial}{\partial r}+\frac{\cos^2\phi}{r^2}\frac{\partial^2}{\partial \phi^2}.}
\end{aligned} \tag{25}$$

(c) (23) 식과 (25) 식으로부터, 다음과 같은 결과를 얻는다.

$$\begin{aligned}\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2} &= \cos^2\phi\frac{\partial^2}{\partial r^2}+\frac{2\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial\phi}-\frac{2\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial r\partial\phi}+\frac{\sin^2\phi}{r}\frac{\partial}{\partial r}+\frac{\sin^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2}\\ &\quad+\sin^2\phi\frac{\partial^2}{\partial r^2}-\frac{2\cos\phi\sin\phi}{r^2}\frac{\partial}{\partial\phi}+\frac{2\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial r\partial\phi}+\frac{\cos^2\phi}{r}\frac{\partial}{\partial r}+\frac{\cos^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2},\\ &= (\sin^2\phi+\cos^2\phi)\frac{\partial^2}{\partial r^2}+\frac{(\sin^2\phi+\cos^2\phi)}{r}\frac{\partial}{\partial r}+\frac{(\sin^2\phi+\cos^2\phi)}{r^2}\frac{\partial^2}{\partial\phi^2},\\ &= \boxed{\frac{\partial^2}{\partial r^2}+\frac{\partial}{r\partial r}+\frac{\partial^2}{r^2\partial\phi^2}.}\end{aligned} \tag{26}$$

3. **(5점)** $\psi_n(x)$ 는 다음과 같다.

$$\begin{aligned}\psi_n(x) &= \sqrt{\frac{1}{2^n n!}}\left(\frac{\mu\omega}{\pi\hbar}\right)^{1/4}H_n\left(\sqrt{\frac{\mu\omega}{\hbar}}x\right)e^{-\mu\omega x^2/(2\hbar)}\\ &= \sqrt{\frac{1}{2^n n!}}\left(\frac{\alpha}{\pi}\right)^{1/4}H_n(\sqrt{\alpha}x)e^{-\alpha x^2/2}\\ &= \left(\frac{1}{2^n n!}\right)^{1/2}\left(\frac{\alpha}{\pi}\right)^{1/4}H_n(y)e^{-y^2/2}\end{aligned} \tag{27}$$

위 식에서

$$\alpha\equiv\frac{\mu\omega}{\hbar}=\left(\frac{\mu k}{\hbar^2}\right)^{1/2},\qquad \text{그리고}\qquad y\equiv\sqrt{\alpha}x \tag{28}$$

그리고 에르미트 다항식 $H_n(y)$ 은 다음과 같은 회귀식이 성립한다.

$$H_{n+1}(y)-2yH_n(y)+2nH_{n-1}(y)=0 \tag{29}$$

(27) 식을 이용하면 $\langle x^2\rangle$ 은 다음과 같이 쓸 수 있다.

$$\langle x^2\rangle = \int_{-\infty}^{\infty}\psi_n*(x)\hat{x^2}\psi_n(x)dx$$

$$= \left(\frac{1}{2^n n!}\right)\left(\frac{\alpha}{\pi}\right)^{1/2}\left(\frac{1}{\alpha}\right)^{3/2}\int_{-\infty}^{\infty}H_n(y)y^2H_n(y)e^{-y^2}dy \tag{30}$$

$$= \left(\frac{1}{2^n n!}\right)\left(\frac{1}{\pi\alpha^2}\right)^{1/2}\int_{-\infty}^{\infty}\left[yH_n(y)\right]^2e^{-y^2}dy \tag{31}$$

(29) 식으로부터 $yH_n(y)=\frac{1}{2}H_{n+1}(y)+nH_{n-1}(y)$ 이므로 이를 (32) 식에 대입하면

$$= \left(\frac{1}{2^n n!}\right)\left(\frac{1}{\pi\alpha^2}\right)^{1/2}\int_{-\infty}^{\infty}\left[\frac{1}{2}H_{n+1}(y)+nH_{n-1}(y)\right]^2e^{-y^2}dy \tag{32}$$

$$\begin{aligned}= \left(\frac{1}{2^n n!}\right)\left(\frac{1}{\pi\alpha^2}\right)^{1/2}&\left[\frac{1}{4}\int_{-\infty}^{\infty}H_{n+1}(y)H_{n+1}(y)e^{-y^2}dy\right.\\ &\left.+n\int_{-\infty}^{\infty}H_{n+1}(y)H_{n-1}(y)e^{-y^2}dy+n^2\int_{-\infty}^{\infty}H_{n-1}(y)H_{n-1}(y)e^{-y^2}dy\right]^2\end{aligned}$$

$\int_{-\infty}^{\infty} H_m(y)H_n(y)e^{-y^2} = \pi^{1/2}2^n n!\delta_{mn}$ 이므로

$$= \left(\frac{1}{2^n n!}\right)\left(\frac{1}{\pi\alpha^2}\right)^{1/2}\left[\frac{\pi^{1/2}2^{n+1}(n+1)!}{4} + n^2\pi^{1/2}2^{n-1}(n-1)!\right] \tag{33}$$

$$= \frac{n+1}{2\alpha} + \frac{n}{2\alpha} = \left(n+\frac{1}{2}\right)\frac{1}{\alpha} = \left(n+\frac{1}{2}\right)\frac{\hbar}{\mu\omega} \tag{34}$$

4. **(총 10점, 각 5점)**

(a) 문제의 Schrödinger 방정식을 다시 쓰면 다음과 같다.

$$\begin{aligned} &-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\psi(x,y) + \left(\frac{1}{2}k_x x^2 + \frac{1}{2}k_y y^2\right)\psi(x,y) = E\psi(x,y) \\ &-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,y,z)}{\partial x^2} - \frac{\hbar^2}{2m}\frac{\partial^2\psi(x,y,z)}{\partial y^2} + \left(\frac{1}{2}k_x x^2 + \frac{1}{2}k_y y^2\right)\psi(x,y) = E\psi(x,y,z) \end{aligned} \tag{35}$$

위 미분방정식의 해가 단일 변수로 이루어진 두 함수의 곱이라고 가정하고, 변수 분리법을 이용하면 $\psi(x,y)$를 다음과 같이 나타낼 수 있다.

$$\psi(x,y) = X(x)Y(y). \tag{36}$$

(36) 식을 (35) 식에 대입하여 정리하면 다음과 같은 결과를 얻는다.

$$-\frac{1}{X(x)}\frac{\hbar^2}{2m}\frac{\partial^2 X(x)}{\partial x^2} + \frac{1}{2}k_x x^2 - \frac{1}{Y(y)}\frac{\hbar^2}{2m}\frac{\partial^2 Y(y)}{\partial y^2} + \frac{1}{2}k_y y^2 = E$$

$$-\frac{1}{X(x)}\frac{\hbar^2}{2m}\frac{\partial^2 X(x)}{\partial x^2} + \frac{1}{2}k_x x^2 = \frac{1}{Y(y)}\frac{\hbar^2}{2m}\frac{\partial^2 Y(y)}{\partial y^2} - \frac{1}{2}k_y y^2 + E \equiv E_x$$

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2X(x)}{dx^2} + \frac{1}{2}k_x x^2 X(x) = E_x X(x)} \tag{37}$$

$$-\frac{1}{Y(y)}\frac{\hbar^2}{2m}\frac{\partial^2 Y(y)}{\partial y^2} + \frac{1}{2}k_x x^2 = E - E_x = E_y \tag{38}$$

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2Y(y)}{dy^2} + \frac{1}{2}k_y y^2 Y(y) = E_y Y(y)} \tag{39}$$

(b) (38) 식으로부터

$$E = E_x + E_y \tag{40}$$

임을 알 수 있고, (37) 식과 (39) 식은 각각 x와 y 방향에서 진동하는 단조화 진동자에 대한 시간비의존 Schrödinger 방정식임을 알 수 있다. (37) 식과 (39) 식의

에너지 고유값은 각각 다음과 같다.

$$E_{x,n_x} = \left(n_x + \frac{1}{2}\right)\hbar\omega_x \qquad (n_x = 0, 1, 2, \cdots) \tag{41}$$

$$E_{x,n_y} = \left(n_y + \frac{1}{2}\right)\hbar\omega_y \qquad (n_y = 0, 1, 2, \cdots) \tag{42}$$

위 식에서 $\omega_x = \sqrt{k_x/m}$ 이고, $\omega_y = \sqrt{k_y/m}$ 이다. (41) 식과 (42) 식을 (40) 식에 대입하면 2차원 단조화 진동운동의 에너지 고유값은 다음과 같다.

$$E_{n_x,n_y} = E_{x,n_x} + E_{y,n_y} = \left(n_x + \frac{1}{2}\right)\hbar\omega_x + \left(n_y + \frac{1}{2}\right)\hbar\omega_y \tag{43}$$
$$(n_x = 0, 1, 2, \cdots \quad \text{그리고} \quad n_y = 0, 1, 2, \cdots)$$

5. **(총 20점, 각 5점)**

(a)

$$\mu_{\mathrm{CO}} = \frac{m_{\mathrm{C}} m_{\mathrm{O}}}{m_{\mathrm{C}} + m_{\mathrm{O}}} = \frac{[(12.00\ \mathrm{g\ mol^{-1}})/N_A][(15.99\ \mathrm{g\ mol^{-1}})/N_A]}{[(12.00\ \mathrm{g\ mol^{-1}})/N_A] + [(15.99\ \mathrm{g\ mol^{-1}})/N_A]}$$
$$= \frac{6.855\ \mathrm{g\ mol^{-1}}}{N_A} = \frac{6.86\ \mathrm{g\ mol^{-1}}}{6.022 \times 10^{23}\ \mathrm{mol^{-1}}} = \mathbf{1.138 \times 10^{-23}}\ \mathrm{g} = \mathbf{1.138 \times 10^{-26}}\ \mathrm{kg} \tag{44}$$

(b)

$$\omega = 2\pi\nu = 2\pi c\bar{\nu} = \left(\frac{k}{\mu}\right)^{1/2}$$
$$\Rightarrow k = (2\pi c\bar{\nu})^2\mu$$
$$= \left[(2\pi)(2.998 \times 10^8\ \mathrm{m\ s^{-1}})(2169.814\ \mathrm{cm^{-1}})\right]^2(1.138 \times 10^{-23}\ \mathrm{g})$$
$$= \left[(2\pi)(2.998 \times 10^8\ \mathrm{m\ s^{-1}})(2169.814 \times 10^2\ \mathrm{m^{-1}})\right]^2(1.138 \times 10^{-26}\ \mathrm{kg})$$
$$= \mathbf{1901}\ \mathrm{N\ m^{-1}} \tag{45}$$

(c) 단조화 진동자의 진동운동에 대하여 $\langle x\rangle = 0$ 이고, (34) 식으로부터 $\langle x^2\rangle = (n + 1/2)\hbar/(\mu\omega)$. 그러므로

$$\delta_x = \left(\langle x^2\rangle - \langle x\rangle^2\right)^{1/2} = \left[\left(n + \frac{1}{2}\right)\frac{\hbar}{\mu\omega}\right]^{1/2} \tag{46}$$

$n = 0$ 이고 $\omega = 2\pi c\bar{\nu}$ 이므로 (46) 식으로부터

$$\delta_x = \left[\frac{1}{2} \times \frac{(1.055 \times 10^{-34}\ \mathrm{J\ s})}{(1.138 \times 10^{-26}\ \mathrm{kg})((2\pi)(2.998 \times 10^8\ \mathrm{m\ s^{-1}})(2169.814\ \mathrm{cm^{-1}})}\right]^{1/2}$$
$$= \mathbf{3.36}\ \mathrm{pm}$$

(d) 흡수하는 전자기 복사에너지와 파장을 각각 $\Delta E_{\rm ph}$ 와 $\lambda_{\rm ph}$ 라 하면,

$$\begin{aligned}\Delta E_{\rm ph} = \frac{hc}{\lambda_{\rm ph}} &= E_f - E_i = \left(n_f + \frac{1}{2}\right)\hbar\omega - \left(n_i + \frac{1}{2}\right)\hbar\omega \\ &= E_1 - E_0 = \frac{3}{2}\hbar\omega - \frac{1}{2}\hbar\omega = \hbar\omega = \left(\frac{h}{2\pi}\right)(2\pi\nu_{\rm CO}) = h\nu_{\rm CO} = \frac{hc}{\lambda_{\rm CO}}\end{aligned} \tag{47}$$

위 식에서 $\lambda_{\rm CO}$ 은 $^{12}{\rm C}^{16}{\rm O}$ 진동운동의 파장이다. (47) 식으로부터

$$\begin{aligned}\lambda_{\rm ph} = \lambda_{\rm CO} = \frac{1}{\bar{\nu}} &= \frac{1}{2169.814\ {\rm cm}^{-1}} \\ &= \mathbf{4.608690 \times 10^{-4}}\ {\rm cm} = \mathbf{4.608690 \times 10^{-6}}\ {\rm m} = \mathbf{4608.690}\ {\rm nm}\end{aligned}$$

위 식으로부터 단조화 진동자의 진동운동의 파장은 흡수하거나 방출하는 전자기 복사 즉, 광자의 파장과 같음을 알 수 있다.

6. **(총 10점, 각 5점)**

(a) (44) 식으로부터 $\mu_{\rm CO} = 1.138 \times 10^{-26}$ kg 이므로

$$\begin{aligned}I_{\rm CO} = \mu r_{eq}^2 &= (1.138 \times 10^{-26}\ {\rm kg})(112.8\ {\rm pm})^2 \\ &= (1.138 \times 10^{-26}\ {\rm kg})(112.8 \times 10^{-12}\ {\rm m})^2 = \mathbf{1.448 \times 10^{-46}}\ {\rm kg\ m^2}\end{aligned} \tag{48}$$

(b) 방출하는 전자기 복사에너지와 파장을 각각 $\Delta E_{\rm ph}$ 와 $\lambda_{\rm ph}$ 라 하면,

$$\begin{aligned}\Delta E_{\rm ph} = \frac{hc}{\lambda_{\rm ph}} &= E_f - E_i = \frac{l_f(l_f+1)\hbar^2}{2I} - \frac{l_i(l_i+1)\hbar^2}{2I} \\ &= E_1 - E_0 = (2-0) \times \frac{\hbar^2}{2I_{\rm CO}}\end{aligned}$$

그러므로

$$\begin{aligned}\Rightarrow \lambda_{\rm ph} &= \frac{2\pi c I_{\rm CO}}{\hbar} \\ &= E_1 - E_0 = \frac{(2\pi)(2.998 \times 10^8\ {\rm m\ s^{-1}})(1.448 \times 10^{-46}\ {\rm kg\ m^2})}{(1.055 \times 10^{-34}\ {\rm J\ s})} = \mathbf{2.59 \times 10^{-3}}\ {\rm m}\end{aligned}$$